

# **L5c:** Introduction to Linear Programming

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Allocation, duality, and resource values  
CHEME 5800 Cornell University

# Objectives for Today

How should we allocate limited resources among competing uses, and how can we show that a feasible allocation is optimal?

## Today's concepts

- **Formulate resource-allocation models:** identify decision variables, objective coefficients, constraints, and bounds in consumer-choice and minimum-cost-flow problems.
- **Explain duality and resource values:** construct the dual, show how feasible dual solutions bound the primal objective, and read the optimal dual variable as the value of an extra dollar of budget.
- **Evaluate optimization results:** distinguish feasible from optimal; check status, recompute the objective, and verify constraints and bounds.

# Lecture and Example

## **Lecture:** [Introduction to Linear Programming](#)

Formulate consumer-choice and network-flow models as linear programs, then use duality to bound the objective and interpret the value of a resource.

## **Example:** [Two Goods Resource Allocation Problem: Apples versus Oranges](#)

Formulate the primal program that maximizes utility within a budget, predict the purchase from utility per dollar, and solve it with the course package.

Optional notebooks on solver algorithms are collected near the end.

# Primal Linear Programming Problems

Suppose a **linear** objective of the continuous decision vector  $\mathbf{x} \in \mathbb{R}^n$  is constrained by  $m$  linear inequalities. The **primal** linear program is:

$$\begin{aligned} & \text{minimize/maximize} && \sum_{i=1}^n c_i x_i \\ & \text{subject to} && \mathbf{A}\mathbf{x} \leq \mathbf{b}, \quad \mathbf{A} \in \mathbb{R}^{m \times n}, \mathbf{b} \in \mathbb{R}^m, \\ & && x_i \geq 0, \quad i = 1, 2, \dots, n. \end{aligned}$$

- $c_i \in \mathbb{R}$ : objective coefficients;  $x_i \in \mathbb{R}$ : decision variables.
- $\mathbf{A}$  and  $\mathbf{b}$ : the constraint matrix and right-hand side vector.

The goal is to minimize or maximize the objective while satisfying the constraints.

# What Does It Mean to Solve This Model?

A decision vector is **feasible** if it satisfies every constraint and bound. All feasible vectors form the **feasible region**.

- An **optimal** vector  $\mathbf{x}^*$  is feasible and attains the best objective value over that region: the largest for a maximization, the smallest for a minimization.
- A feasible vector need not be optimal, and several different vectors can attain the same optimal value.
- If no feasible vector exists, the model is **infeasible**.
- If feasible vectors give arbitrarily good objective values, the problem is **unbounded** and has no finite optimum.

These outcomes depend on the objective, constraints, and bounds together.

# Consumer Choice as a Linear Program

Let there be  $n$  products. For product  $i$ , let  $u_i$  be its utility score and  $p_i$  its price per unit. The consumer has a budget  $I$  and buys  $x_i$  units of product  $i$ , so total cost is  $\sum_i p_i x_i$  and utility is  $U(x_1, \dots, x_n) = \sum_i u_i x_i$ .

The consumer maximizes utility while staying within the budget:

$$\begin{aligned} &\text{maximize} && \sum_{i=1}^n u_i x_i \\ &\text{subject to} && \sum_{i=1}^n p_i x_i \leq I, \\ &&& x_i \geq 0, \quad i = 1, 2, \dots, n. \end{aligned}$$

The optimal solution, if it exists, gives the number of units of each product to purchase. Production planning, transportation, and network flow can be formulated in the same way.

# Case A: Apples versus Oranges

Let  $x_1$  and  $x_2$  be the quantities of apples and oranges. Apples cost 2 dollars per unit and give 0.55 utility units; oranges cost 4 dollars per unit and give 0.45 utility units.

The budget is 100 dollars:

$$\begin{aligned} &\underset{x_1, x_2}{\text{maximize}} && U(x_1, x_2) = 0.55x_1 + 0.45x_2 \\ &\text{subject to} && 2x_1 + 4x_2 \leq 100, \quad x_1, x_2 \geq 0. \end{aligned}$$

Spending the whole budget on one fruit sets a finite upper bound  $x_i \leq I/p_i$ : at most **50 apples** or **25 oranges**.

Utility per dollar decides which fruit we buy:

$$\frac{u_1}{p_1} = \frac{0.55}{2} = \mathbf{0.275}, \quad \frac{u_2}{p_2} = \frac{0.45}{4} = 0.1125.$$

Apples win, so the optimum is  $(x_1^*, x_2^*) = (50, 0)$  with  $U^* = 27.5$ .

# Three Utility Settings, Three Solution Cases

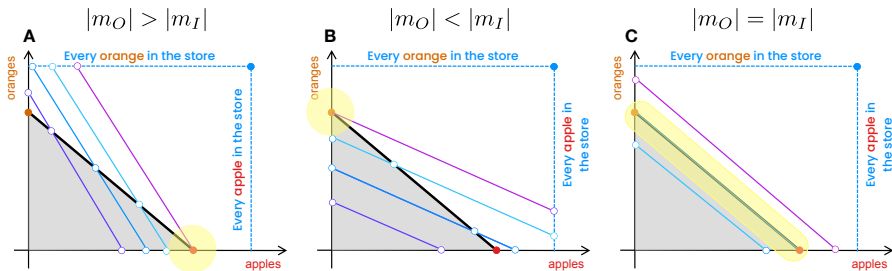
Keep prices  $(p_1, p_2) = (2, 4)$  and budget  $I = 100$  fixed. The data cell offers three utility settings:

| Case | $(u_1, u_2)$ | $(u_1/p_1, u_2/p_2)$ | Optimal purchase      | Utility |
|------|--------------|----------------------|-----------------------|---------|
| A    | (0.55, 0.45) | (0.275, 0.1125)      | (50, 0), apples only  | 27.5    |
| B    | (0.15, 0.55) | (0.075, 0.1375)      | (0, 25), oranges only | 13.75   |
| C    | (2, 4)       | (1, 1)               | Any full-budget mix   | 100     |

The fruit we buy depends on utility **per dollar**, not on utility alone. With linear utility we typically get a **corner solution**.

In the special Case C the ratios are equal, so every mixture that spends the full budget is optimal: infinitely many solutions share the same utility.

# The Optimum Is a Corner or an Entire Edge



Compare the slope of the budget line,  $m_I = -p_1/p_2$ , with the slope of a line of constant utility,  $m_O = -u_1/u_2$ :

- **Case A:**  $|m_I| < |m_O|$  gives the apple corner.
- **Case B:**  $|m_I| > |m_O|$  gives the orange corner.
- **Case C:**  $|m_I| = |m_O|$  makes the whole budget edge optimal.

# Does *E. coli* Have Free Will?

A bacterium must allocate its limited proteome among competing enzymes to maximize growth. This is a consumer-choice problem:

- **Budget:** the total protein the cell can devote to enzymes.
- **Products:** the sugars it can metabolize; each enzyme returns growth per unit of protein invested.
- **Objective:** maximize growth rate within the proteome budget.

Because the utility is linear, the optimum is a **corner**: the cell commits its budget to the single most profitable sugar and consumes sugars in sequence. Only equally profitable sugars are used together, along the budget line.

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**Reading:** [Rethinking the Choice Behavior of Sugar Metabolism in Bacteria \(arXiv\)](#)

# Minimum-Cost Network Flow

A directed graph  $G = (\mathcal{V}, \mathcal{E})$  has source, sink, and intermediate nodes. Edges carry goods or resources between nodes.

- Each edge  $j \in \mathcal{E}$  has a capacity  $c_j$ , the most flow it can carry, and a cost  $w_j$  per unit of flow.
- $f_j$  is the flow on edge  $j$ ;  $s$  is the source and  $t$  the sink.
- The goal: deliver a specified amount of flow from  $s$  to  $t$  at **minimum total cost**.

Several routings can deliver the same amount; their costs differ.

# An Incidence Matrix Encodes the Graph

The matrix  $\mathbf{A} \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{E}|}$  has one row per node  $i$  and one column per edge  $j$ . Its entries and the right-hand side  $b_i$  are:

$$A_{ij} = \begin{cases} 1, & \text{edge } j \text{ is incoming to } i, \\ -1, & \text{edge } j \text{ is outgoing from } i, \\ 0, & \text{otherwise.} \end{cases} \quad b_i = \begin{cases} -F, & i = s \text{ (source),} \\ F, & i = t \text{ (sink),} \\ 0, & \text{otherwise.} \end{cases}$$

Here  $F$  is the total flow from source to sink. The source generates flow, the sink consumes it, and every other node conserves it:

$$\underbrace{(\mathbf{A}\mathbf{f})_i}_{\text{inflow minus outflow}} = b_i.$$

# The Minimum-Cost Flow Linear Program

Collect the edge flows in  $\mathbf{f} \in \mathbb{R}^{|\mathcal{E}|}$  and the net inflows in  $\mathbf{b} \in \mathbb{R}^{|\mathcal{V}|}$ :

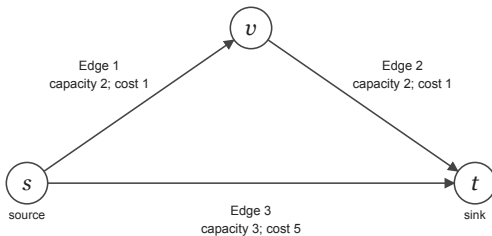
$$\begin{aligned} & \text{minimize} && \sum_{j \in \mathcal{E}} w_j f_j \\ & \text{subject to} && \mathbf{A}\mathbf{f} = \mathbf{b}, \\ & && 0 \leq f_j \leq c_j, \quad \forall j \in \mathcal{E}. \end{aligned}$$

- The objective adds the cost incurred on every edge.
- The equalities enforce the required delivery and flow conservation.
- The bounds keep every edge flow nonnegative and within capacity.

The optimal solution, if it exists, is the cheapest routing that delivers  $F$ .

# Worked Network: Two Routes

Deliver  $F = 3$  flow units from  $s$  to  $t$ . Edge labels give capacity in flow units and cost in dollars per flow unit.



Through  $v$ : cost  $1 + 1 = 2$  dollars per unit, but capacity 2. Direct edge 3: cost 5 dollars per unit.

Some of the 3-unit delivery must travel over edge 3.

# Write the Three Node Balances

Order rows as  $(s, v, t)$  and columns as edges  $(1, 2, 3)$ . For  $F = 3$ , the equality constraints  $\mathbf{A}\mathbf{f} = \mathbf{b}$  are:

$$\underbrace{\begin{bmatrix} -1 & 0 & -1 \\ 1 & -1 & 0 \\ 0 & 1 & 1 \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} f_1 \\ f_2 \\ f_3 \end{bmatrix}}_{\mathbf{f}} = \underbrace{\begin{bmatrix} -3 \\ 0 \\ 3 \end{bmatrix}}_{\mathbf{b}}.$$

- Source:  $f_1 + f_3 = 3$  units leave.
- Intermediate vertex:  $f_1 = f_2$ .
- Sink:  $f_2 + f_3 = 3$  units enter.

Each edge column has a  $-1$  where the edge begins and a  $+1$  where it ends.

# The Program With the Numbers Filled In

With costs  $\mathbf{w} = (1, 1, 5)$  and capacities  $\mathbf{c} = (2, 2, 3)$ , the primal linear program for this network is:

$$\begin{array}{ll}\text{minimize} & f_1 + f_2 + 5f_3 \\ \text{subject to} & f_1 + f_3 = 3, \\ & f_1 - f_2 = 0, \\ & f_2 + f_3 = 3, \\ & 0 \leq f_1 \leq 2, \quad 0 \leq f_2 \leq 2, \quad 0 \leq f_3 \leq 3.\end{array}$$

This is the general model with numbers substituted: the objective is  $\sum_j w_j f_j$ , the three equalities are the rows of  $\mathbf{A}\mathbf{f} = \mathbf{b}$ , and the bounds are  $0 \leq f_j \leq c_j$ .

# Solve by Hand: Fill the Cheaper Route First

The equalities leave one free variable. Let  $q$  be the flow along the route through  $v$ :  $f_1 = f_2 = q$  and  $f_3 = 3 - q$ , with  $0 \leq q \leq 2$  from the bounds on edges 1 and 2.

Substitute into the total cost:

$$C(q) = \sum_{j \in \mathcal{E}} w_j f_j = (1)(q) + (1)(q) + (5)(3 - q) = 15 - 3q.$$

Cost decreases with  $q$ , so use the full capacity of the cheaper route:

$$(f_1^*, f_2^*, f_3^*) = (2, 2, 1), \quad C^* = 9 \text{ dollars.}$$

Sending all 3 units directly over edge 3 is feasible but costs 15 dollars. Conservation and capacity checks establish feasibility; the cost comparison selects the preferred routing.

# What Does This Small Example Tell Us?

- **The solver does the search.** A solver searches the feasible set defined by the constraints and bounds for the point that minimizes the objective. Here that set was a line segment, so we searched by hand; larger networks need a solver.
- **The required flow is data, not a decision.** Asking for more than the network can carry, here more than the 5 units that can leave the source, makes the linear program infeasible.
- **The same formulation scales.** The [L5d lab](#) uses this incidence convention for a worker–task assignment model and examines what changes when an assignment becomes unavailable.

Formulating the model is our job; finding the best point in the feasible set is the solver's.

# Dual Linear Programming Problems

The primal returns a purchase. How do we know that no other feasible purchase does better?

In the [L5b lab](#), a cut answered this question for flows: every feasible flow had to cross the cut, so the cut capacity bounded every flow. We want a similar check for a linear program.

**The idea: put a price on the budget.** Let  $y \geq 0$  be the value of one dollar of budget, in utility units per dollar. Product  $i$  costs  $p_i$  dollars and delivers  $u_i$  utility units, so it returns  $u_i/p_i$  utility per dollar.

If  $y$  is at least the return of every product, so that  $p_i y \geq u_i$  for all  $i$ , then no way of spending  $I$  dollars can earn more than  $Iy$  utility units.

# Why Does This Give a Bound?

Take any feasible purchase  $\mathbf{x} \geq \mathbf{0}$  and any  $y \geq 0$  with  $p_i y \geq u_i$  for every product. Replace each utility by the larger quantity  $p_i y$ , then use the budget constraint:

$$U(\mathbf{x}) = \sum_{i=1}^n u_i x_i \leq \sum_{i=1}^n (p_i y) x_i = y \sum_{i=1}^n p_i x_i \leq yI.$$

- First inequality:  $u_i \leq p_i y$  and  $x_i \geq 0$ .
- Last inequality:  $\sum_i p_i x_i \leq I$  and  $y \geq 0$ .

Any such  $y$  gives an upper bound: no feasible purchase earns more than  $Iy$  utility units.

# The Dual Finds the Smallest Upper Bound

Different valid values of  $y$  give different bounds; the smallest is the most informative. Minimizing  $Iy$  over valid  $y$  is the **consumer dual problem**:

$$\begin{aligned} & \underset{y}{\text{minimize}} && Iy \\ & \text{subject to} && p_i y \geq u_i, \quad i = 1, \dots, n, \\ & && y \geq 0. \end{aligned}$$

The constraints say  $y \geq u_i/p_i$  for every product, so the smallest feasible  $y$  is the best return available:

$$y^* = \max_i \frac{u_i}{p_i},$$

the product with the **biggest bang for the buck**. The optimal dual value  $Iy^*$  is the best upper bound on utility, and any feasible purchase that attains it is optimal.

# What Is the Consumer's Budget Worth?

Take Case A: apples cost 2 dollars and give 0.55 utility units per unit, oranges cost 4 dollars and give 0.45, and  $I = 100$  dollars. The dual constraints are  $2y \geq 0.55$  and  $4y \geq 0.45$ , so:

$$y^* = \max\left\{\frac{0.55}{2}, \frac{0.45}{4}\right\} = 0.275, \quad Iy^* = 27.5.$$

Buying 50 apples and no oranges is feasible (it spends exactly 100 dollars) and attains  $0.55 \cdot 50 = 27.5$  utility units.

That equals the dual bound  $Iy^* = 27.5$ , so no feasible purchase can do better: this purchase is **optimal**.

# Both Inequalities Hold With Equality

Substitute  $\mathbf{x} = (50, 0)$  and  $y^* = 0.275$  into the chain:

$$\begin{aligned} \underbrace{0.55 \cdot 50 + 0.45 \cdot 0}_{U(\mathbf{x})=27.5} &\leq \underbrace{(2 \cdot 0.275) \cdot 50 + (4 \cdot 0.275) \cdot 0}_{=27.5} \\ &= \underbrace{0.275 \cdot (2 \cdot 50 + 4 \cdot 0)}_{=27.5} \leq \underbrace{0.275 \cdot 100}_{Iy^*=27.5}. \end{aligned}$$

The first holds with equality because every unit bought (apples) has  $p_i y^* = u_i$  exactly; the second because the purchase spends the whole budget.

**Shadow price.**  $y^* = 0.275$  is the budget's **shadow price** in utility units per dollar: an additional dollar buys 0.5 units of apples and increases optimal utility by 0.275.

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**Assumes:** fixed prices and utilities, divisible quantities, and sufficient availability. In general models, shadow prices can change as resource availability changes.

# The General Primal–Dual Pair

## Primal problem

$$\begin{aligned} &\text{maximize} && \sum_{i=1}^n c_i x_i \\ &\text{s.t.} && \sum_{i=1}^n A_{j,i} x_i \leq b_j, \quad j = 1, \dots, m, \\ &&& x_i \geq 0, \quad i = 1, \dots, n. \end{aligned}$$

## Dual problem

$$\begin{aligned} &\text{minimize} && \sum_{j=1}^m b_j y_j \\ &\text{s.t.} && \sum_{j=1}^m A_{j,i} y_j \geq c_i, \quad i = 1, \dots, n, \\ &&& y_j \geq 0, \quad j = 1, \dots, m. \end{aligned}$$

The consumer problem is this pair with a single constraint,  $m = 1$ : the coefficients  $c_i$  are the utilities  $u_i$ , the one constraint row  $A_{1,i}$  holds the prices  $p_i$ , the right-hand side  $b_1$  is the budget  $I$ , and the lone dual variable  $y_1$  is the budget value  $y$ .

# What Has Changed Between Primal and Dual?

With  $\mathbf{A} \in \mathbb{R}^{m \times n}$ ,  $\mathbf{c} \in \mathbb{R}^n$ , and  $\mathbf{b} \in \mathbb{R}^m$  as above:

1. The **sense of the objective flips**: a maximization primal has a minimization dual. A minimization primal with  $\mathbf{Ax} \geq \mathbf{b}$ ,  $\mathbf{x} \geq \mathbf{0}$  has the maximization dual  $\mathbf{A}^\top \mathbf{y} \leq \mathbf{c}$ ,  $\mathbf{y} \geq \mathbf{0}$ .
2. Primal objective coefficients  $c_i$  become the dual right-hand side.
3. Primal right-hand side constants  $b_j$  become the dual objective coefficients.
4. The constraint matrix is **transposed**:  $\sum_{j=1}^m A_{j,i} y_j$  is component  $i$  of  $\mathbf{A}^\top \mathbf{y}$ .
5. Variables and constraints **swap counts**: the primal has  $n$  variables and  $m$  constraints; the dual has  $m$  variables and  $n$  constraints.

# Constraints and Variables Are Paired, With Signs

Each primal constraint  $j$  gives a dual variable  $y_j$ , and each primal variable  $x_i$  gives a dual constraint  $i$ :

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Primal condition

Corresponding dual condition

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$$\sum_i A_{j,i} x_i \leq b_j$$

$$y_j \geq 0$$

$$\sum_i A_{j,i} x_i = b_j$$

$y_j$  free (no sign restriction)

$$x_i \geq 0$$

$$\sum_j A_{j,i} y_j \geq c_i$$

$x_i$  free

$$\sum_j A_{j,i} y_j = c_i$$

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The network-flow model uses equality balances  $\mathbf{A}\mathbf{f} = \mathbf{b}$ , so its dual variables are free.

# Weak and Strong Duality

For the maximization primal and its dual:

$$\max\{\mathbf{c}^\top \mathbf{x} : \mathbf{Ax} \leq \mathbf{b}, \mathbf{x} \geq \mathbf{0}\} \quad \text{and} \quad \min\{\mathbf{b}^\top \mathbf{y} : \mathbf{A}^\top \mathbf{y} \geq \mathbf{c}, \mathbf{y} \geq \mathbf{0}\}.$$

- **Weak duality.** For any primal-feasible  $\mathbf{x}$  and dual-feasible  $\mathbf{y}$ ,  $\mathbf{c}^\top \mathbf{x} \leq \mathbf{b}^\top \mathbf{y}$ . The primal optimum is bounded above by the dual optimum; the difference is the **duality gap**.
- **Strong duality.** If both problems are feasible with finite optimal values, then  $\max\{\mathbf{c}^\top \mathbf{x}\} = \min\{\mathbf{b}^\top \mathbf{y}\}$ , so the duality gap is zero.

A feasible primal candidate whose objective matches a feasible dual bound is optimal, as the 50-apple purchase was.

# How a Solver Fits the Modeling Workflow

We supply variables, an objective, constraints, and bounds; the solver reports why it stopped and, when available, a candidate. Before interpreting it:

1. **Read the status.** An optimal status means optimal within numerical tolerances. A time or iteration limit may leave a feasible candidate without proving optimality; an infeasible or unbounded model has no finite optimum.
2. **Verify feasibility.** Substitute into the original constraints and bounds: expenditure versus budget, or every node balance in  $\mathbf{A}\mathbf{f} = \mathbf{b}$  and every edge bound.
3. **Recompute the objective.** Use the returned quantities and the original coefficients; compare with the reported value.
4. **Check an optimality bound.** Verify dual feasibility, then compare  $\mathbf{b}^\top \mathbf{y}$  with  $\mathbf{c}^\top \mathbf{x}$ . A zero gap proves optimality in exact arithmetic.

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**Tolerances:** account for the units and scales of the quantities compared. See the [JuMP solution guide](#).

# Optional: How a Solver Finds an Optimum

Three optional notebooks, not prerequisites for the L5d lab, read in order:

- **Lagrange multipliers and KKT conditions.** Why can't we just use Lagrange multipliers on a linear program? Adding multipliers for the nonnegativity bounds gives the Karush–Kuhn–Tucker conditions, necessary and sufficient for optimality.
- **Revised simplex.** How does a solver move from one corner to a better one? Basic and nonbasic variables, reduced costs, and the ratio test.
- **Interior-point methods.** What if we cross the interior instead of walking the edges? A logarithmic barrier, perturbed KKT residuals, and Newton steps.

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**Lab L5d:** [formulate and solve a worker–task minimum-cost flow with JuMP and GLPK.](#)

# Summary

We developed linear programs for consumer choice and network flow, then used duality to bound the objective and value a limited resource.

## Key takeaways

- **Formulating allocation models.** We expressed purchases and flows using continuous variables, linear objectives, and resource or balance constraints.
- **Interpreting duality and resource values.** We used feasible dual solutions to bound the primal objective and read optimal dual variables as shadow prices, the marginal value of a resource.
- **Evaluating optimization results.** We distinguished feasibility from optimality, checked status, constraints, bounds, and a recomputed objective, and proved optimality by matching primal and dual values.

The primal asks how best to use the resources; the dual bounds what they can achieve. Reaching that bound justifies optimality.